values was used instead. We abstained from making any revisions to the initial forecasts. We did so because determining when to update and what information to employ for this update would have necessitated a significant degree of human involvement in the process, which we sought to minimize given that updating is a crucial skill for human forecasters.[23] Since repeated updating is fundamental to the method of crowd-aggregated forecasting, it was important to account for GPT-4's inability to autonomously carry out this task in order to enable unbiased inferences from our results.

In order to control for these confounding factors, our design entailed recording both the initial model forecast and the crowd-aggregated median forecast at the close of the same day at which the model forecast was submitted. This allowed us to control for updating over extended question periods, which is done by humans, but is difficult to operationalize for GPT-4 without introducing bias in deciding when and how to prompt the update. Our comparison thus directly put GPT-4 on par with the human forecasters by only looking at initial forecasts. The full data encompassed for each question GPT-4's forecast, the median human-crowd forecast at the conclusion of the same day, the ground truth, whether a community prediction was visible at the time of prediction, the number of human forecasters at this point in time (for a median number of 37 forecasters), and the total duration, spanning the timeframe from when predictions were made to the eventual resolution of the question. This set of data was collected for all 23 binary forecasts.

3 Results

First, we examine the probabilistic forecasts of both GPT-4 and the human crowd for each question in a directional analysis. See Figure 1 for a paired comparison plot illustrating the respective probability forecasts, with the upper panel displaying questions that resolved positively and the lower panel displaying questions that resolved negatively. These data indicate that in 18 out of 23 questions, the median human-crowd forecasts were directionally closer to the truth than GPT-4's predictions, $\chi^2(1) = 12.52, p = .001$.

We also compared the frequency with which each of GPT-4's forecast and the aggregate human forecast was directionally correct, i.e., on the correct side of the probability midpoint. When doing so, we observed that the GPT-4 forecast was directionally correct 69.57% of the time, in contrast to 95.65% of the time for the aggregate human forecast, although the difference in distributions between the two was not statistically significant, $\chi^2(1) = 3.78, p = .052$. Similarly, the 69.57% proportion of directionally correct answers for GPT-4 also did not statistically deviate from the random 50% baseline, $Z = 1.88, p = .061$. These data suggest that while the aggregate human forecasts were comparatively more often closer to the truth than those of GPT-4, we do not find statistically significant results when testing this independently in relation to the probability scale midpoint.

The primary pre-registered outcome of interest for our analysis is forecasting accuracy. To compute this accuracy as our dependent variable, we employ Brier scores,[5] with the score for each individual provided below. Here, f is the forecasted probability and o is the observed outcome (which is 1 if the event occurs and 0 if it does not), given by $B = (f - o)^2$. We then aggregate the question-level Brier scores per condition as

$$\frac{1}{N}\sum_{i=1}^{N}(f_{i,\text{condition}} - o_i)^2. \tag{1}$$

A perfect accuracy would yield a Brier score of 0, while perfect inaccuracy would result in a Brier score of 1.

In our data, we calculate the mean accuracy by aggregating the the question-level Brier scores for GPT-4's predictions and the median human-crowd forecasts. We observe an average Brier score for GPT-4's predictions of $B = .20$ ($SD = .18$), while the human forecaster average Brier score was $B = .07$ ($SD = .08$). Initially, we test both GPT-4's and the human crowd's accuracy against a simple no-information baseline. This baseline is the Brier score of 0.25, equivalent to predicting 50% on each question. This serves as a first test of prediction accuracy. Our analysis reveals that GPT-4 does not exhibit predictive performance that is statistically distinct from the no-information baseline, $t(22) = -1.23, p = 0.23$. On the other hand, we find that the human crowd's accuracy is significantly superior to that of the no-information baseline, $t(22) = -10.69, p < .001$. These

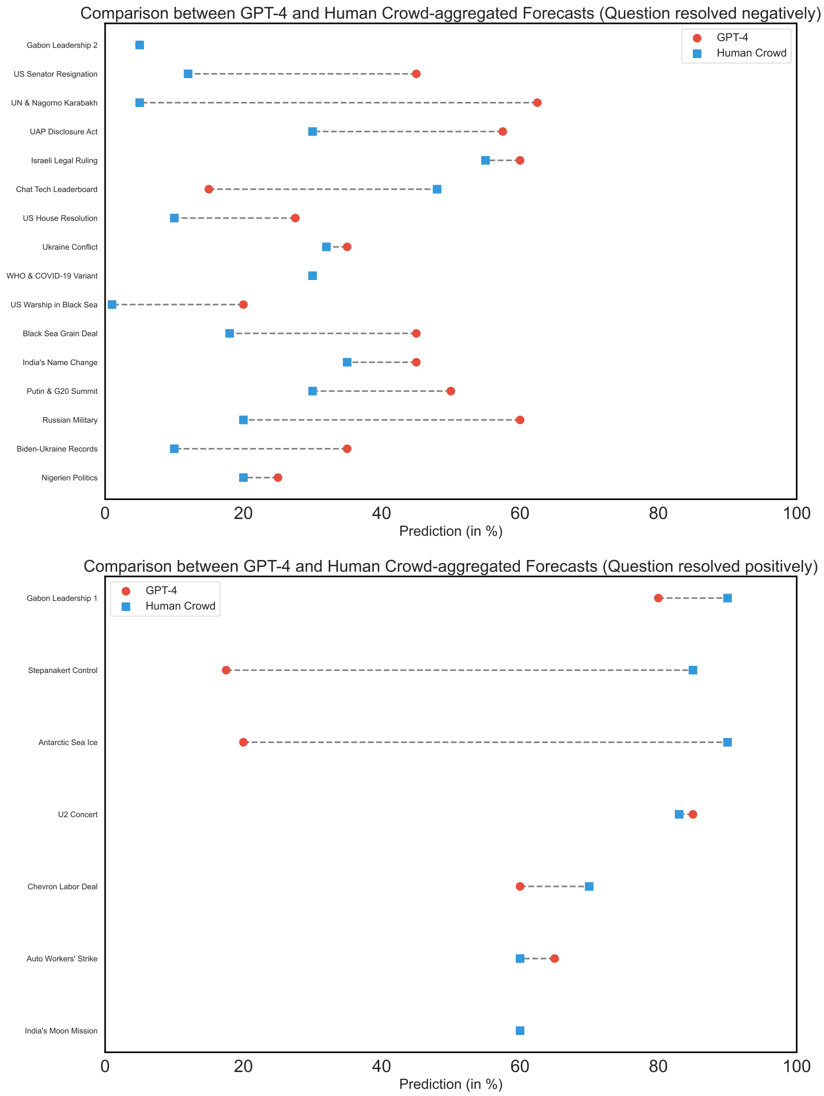

Figure 1: Paired comparison plot showing probability forecasts (in %) per question for the median human-crowd forecast and the GPT-4 forecast. The top panel lists questions that resolved negatively. The bottom panel lists questions that resolved positively.

findings suggest that while the aggregate human-crowd forecasts have high accuracy, the mean LLM forecasts do not significantly improve on the no-information baseline.

In order to test our pre-registered null hypothesis, we compare the mean accuracy of GPT-4 and that of the human crowd. This yields a total mean difference of $\Delta = .13$, which is statistically significant at the two-tailed test, $t(44) = 3.11, p = .003$, with an effect size of Cohen's $d = .94$. Based on this, we reject our null hypothesis, and conclude that GPT-4 significantly underperforms in this real-world forecasting tournament compared to the median human-crowd forecasts that are currently employed in forecasting tournaments. We also find that GPT-4's accuracy was not significantly different on questions that included the community prediction compared to those that did not, $t(21) = -1.81, p = 0.085$, suggesting that this result is explained by an internal factor of GPT-4's forecasting performance. For a graphical depiction of the distribution of Brier scores per condition, see Figure 2.

One potential reason that might explain this outcome could be that GPT-4's predictions, due to its reinforcement learning from human feedback,[40] may gravitate towards the probability scale

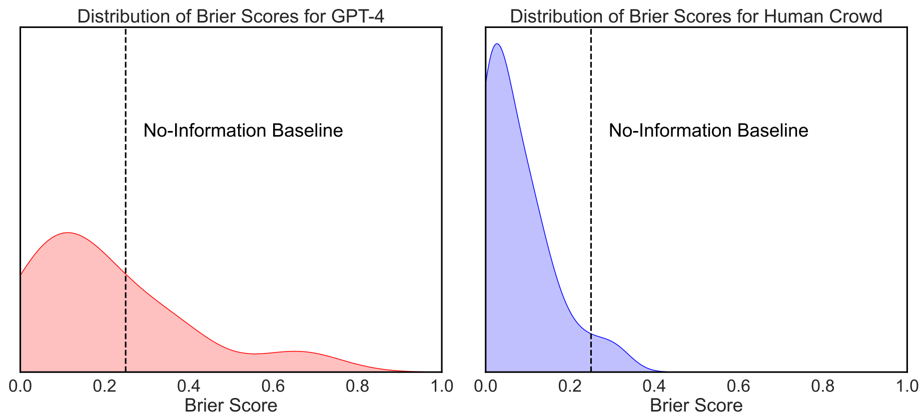

Figure 2: Kernel Density Estimation (KDE) plots of Brier scores for GPT-4's forecasts and median human-crowd forecasts. Black dotted line represents a Brier score of 0.250 for the 'No-Information Baseline'.

midpoint (50%) in a form of model conservatism. This could explain the low prediction accuracy in our forecasting tournament with a small-to-medium sample size of questions. To analyse this, we conducted the following exploratory analyses. First, we computed the coefficient of variation (CV) as a normalized measure of dispersion for the predictions made by both GPT-4 and the human crowd. Our analysis found a CV of 48.22% for GPT-4 and 73.67% for the human crowd when centered on the mean, suggesting that GPT-4 has less dispersion. Similarly, when anchoring the CV around the midpoint of the probability scale, the values were 42.14% for GPT-4 and 57.59% for the human crowd respectively, again showing the pattern of GPT-4 having less dispersion.

These analyses might suggest that GPT-4's predictions could be more densely clustered around both the mean and the midpoint compared to the human crowd, aligning with a more cautious prediction pattern: one that is less inclined towards extreme forecasts. However, our inferential analysis employing Levene's test for equality of variances at the 50% midpoint did not yield statistically significant evidence to substantiate differences in variance around the midpoint, $F(1, 44) = .54, p = .47$. Thus, our results do not provide grounds for asserting a difference in variance around the probability scale midpoint. See Figure 3 for a graphical representation of these findings.

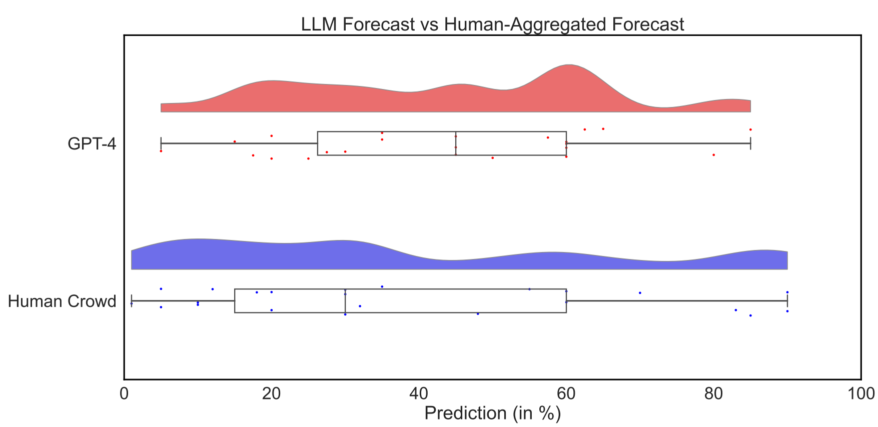

Figure 3: Raincloud plot of the distribution of probability forecasts (in %) made by GPT-4 and by the median human-crowd forecast for all questions. Box plots represent interquartile ranges.

Furthermore, in an additional exploratory analysis, we probe whether the relationship between the duration of a forecasting question remaining unresolved and the forecasting accuracy at the inception of this period significantly diverges between GPT-4's forecasts and human-crowd forecasts. This